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Grouped with:

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ENGG 27.01 - M

March 16, 2026

Project 2: Power Method for Eigenvalues and Eigenvectors

```
testMatrix
-8.8   0.5   6.2   1.3   -7.2   comments
9.6    6.7   -1    -0.8  -6.4   comments
-4.3   7.3   2.2   -0.7  3.1    comments
-9.4   -7.7  -0.2  -8.4  -4.4   comments
-7.7   2.5   -3    -6.4  4.2    comments
comments
```

sample screenshot – testMatrix file used for testing

```
PS D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 27 - 2\Project 2> ./main
Hello, welcome to find the Eigenvalue and Eigenvector-inator!
Please input the name of a matrix file like 'testMatrix': testMatrix
Matrix file 'testMatrix' opened successfully.
Success! Detected a square 5x5 matrix.
Display trace of intermediate estimates? (0 for no, 1 for yes): 1
==========================================
Test Unit Vector: [1, 0, 0, 0, 0] transpose
------------------------------------------
[0]     Eigenvalue = 9.6
        Eigenvector: -0.916667  |  1  |  -0.447917  |  -0.979167  |  -0.802083  |
------------------------------------------
[1]     Eigenvalue = 13.8
        Eigenvector: 0.745773  |  0.309028  |  0.612696  |  0.924668  |  1  |
------------------------------------------
[2]     Eigenvalue = -21.6795
        Eigenvector: 0.397034  |  -0.0681507  |  -0.131449  |  1  |  0.393267  |
------------------------------------------
[3]     Eigenvalue = -13.3114
        Eigenvector: 0.44131  |  -0.0127298  |  0.148354  |  1  |  0.569545  |
------------------------------------------
```

```
------------------------------------------
[56]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[57]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[58]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[59]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
Eigenvalues and Eigenvector converged.
------------------------------------------
Eigenvalue: -14.348097234551389
Eigenvector:
0.37843325754247736
0.020798125181844686
0.036802278566342439
1
0.50529968046332463
==========================================
```

sample screenshots – program run, first four and last four iterations of first unit vector results

```
==========================================
Test Unit Vector: [0, 1, 0, 0, 0] transpose
------------------------------------------
[0]     Eigenvalue = -7.7
        Eigenvector: -0.0649351  |  -0.87013  |  -0.948052  |  1  |  -0.324675  |
------------------------------------------
[1]     Eigenvalue = -9.86494
        Eigenvector: 0.21327  |  0.428515  |  1  |  -0.0535808  |  0.66851  |
------------------------------------------
[2]     Eigenvalue = -7.99567
        Eigenvector: 0.043204  |  0.0396657  |  -0.815565  |  1  |  0.0525577  |
------------------------------------------
[3]     Eigenvalue = -9.17968
        Eigenvector: 0.489699  |  -0.0391858  |  0.242659  |  1  |  0.432049  |
------------------------------------------
```

```
------------------------------------------
[60]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[61]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[62]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[63]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
Eigenvalues and Eigenvector converged.
------------------------------------------
Eigenvalue: -14.348097234551389
Eigenvector:
0.37843325754247736
0.020798125181844686
0.036802278566342439
1
0.50529968046332463
==========================================
```

sample screenshots – first four and last four iterations of second unit vector results

```
==========================================
Test Unit Vector: [0, 0, 1, 0, 0] transpose
------------------------------------------
[0]     Eigenvalue = 6.2
        Eigenvector: 1  |  -0.16129  |  0.354839  |  -0.0322581  |  -0.483871  |
------------------------------------------
[1]     Eigenvalue = 11.2871
        Eigenvector: -0.286939  |  1  |  -0.547013  |  -0.516433  |  -0.973993  |
------------------------------------------
[2]     Eigenvalue = 11.1391
        Eigenvector: 0.536396  |  1  |  0.419472  |  0.334878  |  0.499581  |
------------------------------------------
[3]     Eigenvalue = -17.8371
        Eigenvector: 0.268047  |  -0.446524  |  -0.405369  |  1  |  0.164468  |
------------------------------------------
```

```
------------------------------------------
[61]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[62]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[63]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[64]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
Eigenvalues and Eigenvector converged.
------------------------------------------
Eigenvalue: -14.348097234551389
Eigenvector:
0.37843325754247736
0.020798125181844686
0.036802278566342439
1
0.50529968046332463
==========================================
```

sample screenshots – first four and last four iterations of third unit vector results

```
==========================================
Test Unit Vector: [0, 0, 0, 1, 0] transpose
------------------------------------------
[0]     Eigenvalue = -8.4
        Eigenvector: -0.154762  |  0.0952381  |  0.0833333  |  1  |  0.761905  |
------------------------------------------
[1]     Eigenvalue = -11.0476
        Eigenvector: 0.204526  |  0.59806  |  -0.290194  |  1  |  0.182866  |
------------------------------------------
[2]     Eigenvalue = -15.6742
        Eigenvector: 0.211599  |  -0.273718  |  -0.173204  |  1  |  0.308857  |
------------------------------------------
[3]     Eigenvalue = -9.60573
        Eigenvector: 0.41606  |  0.25048  |  0.315604  |  1  |  0.717987  |
------------------------------------------
```

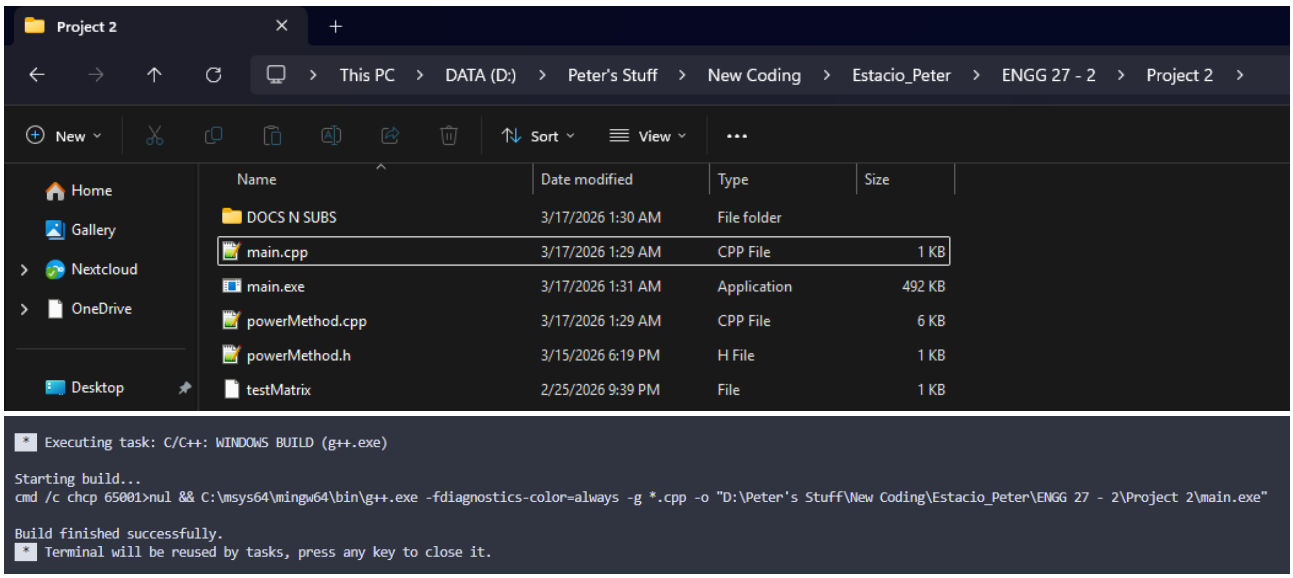
```
------------------------------------------
[58]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[59]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[60]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[61]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
Eigenvalues and Eigenvector converged.
------------------------------------------
Eigenvalue: -14.348097234551389
Eigenvector:
0.37843325754247736
0.020798125181844686
0.036802278566342439
1
0.50529968046332463
==========================================
```

sample screenshots – first four and last four iterations of fourth unit vector results

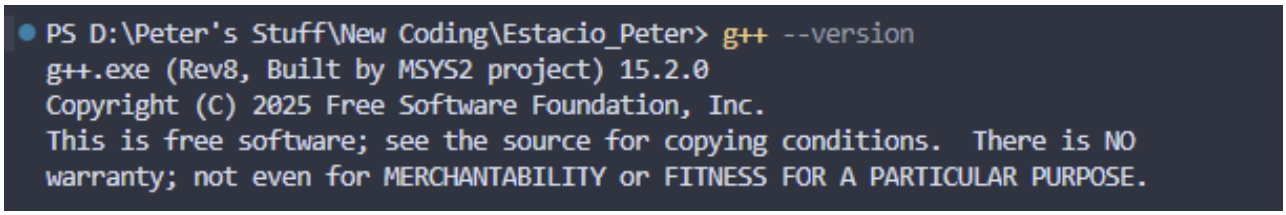
```
==========================================
Test Unit Vector: [0, 0, 0, 0, 1] transpose
------------------------------------------
[0]     Eigenvalue = -7.2
        Eigenvector: 1  |  0.888889  |  -0.430556  |  0.611111  |  -0.583333  |
------------------------------------------
[1]     Eigenvalue = 19.2306
        Eigenvector: -0.313592  |  1  |  -0.0517117  |  -0.973711  |  -0.548462  |
------------------------------------------
[2]     Eigenvalue = 8.99801
        Eigenvector: 0.624816  |  0.892459  |  0.835301  |  0.650201  |  1  |
------------------------------------------
[3]     Eigenvalue = -22.774
        Eigenvector: 0.273471  |  -0.185398  |  -0.364924  |  1  |  0.221618  |
------------------------------------------
```

```
------------------------------------------
[61]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[62]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[63]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
[64]    Eigenvalue = -14.3481
        Eigenvector: 0.378433  |  0.0207981  |  0.0368023  |  1  |  0.5053  |
------------------------------------------
Eigenvalues and Eigenvector converged.
------------------------------------------
Eigenvalue: -14.348097234551389
Eigenvector:
0.37843325754247736
0.020798125181844686
0.036802278566342439
1
0.50529968046332463
PS D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 27 - 2\Project 2>
```

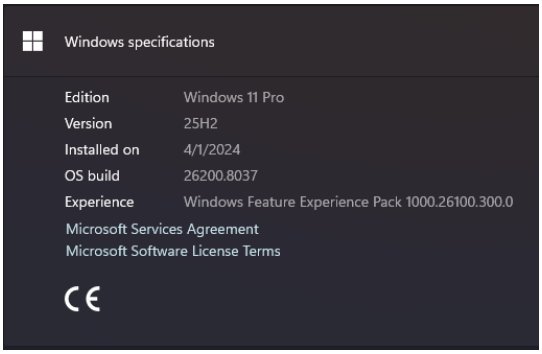
sample screenshots – first four and last four iterations of fifth unit vector results



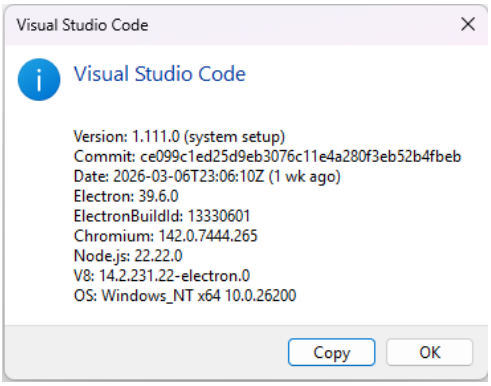
sample screenshot – compilation on Visual Studio Code



compiler version – g++ version 15.2.0 (Rev8, Built by MSYS2 project)



operating system - Windows 11 Pro, 25H2, build 26200.8037



IDE – Visual Studio Code ver 1.111.0

Project Evaluation - Power Method for Eigenvalues and Eigenvectors		
Item	**Points**	**Rubrics**
early work	8/8	(all or nothing) 8: at most 10% of the code in the final implementation differs from that in early work submission
implementation and testing environment report	4/4	4 - all instructions followed correctly
user interface: input file	4/4	4 - appropriate user interface provided
parsing of matrix files	8/8	8 - application properly parses matrix files
documentation of Power Method implementation	32/32	32 - properly working and clearly documented implementation of the Power Method
convergence	8/8	16 - all instructions related to convergence are properly implemented
divergence	8/8	16 - divergence is always detected properly
generality	16/16	16 - implementation of Power Method works for all cases
matrix file	4/4	4 - correctly prepared matrix file
program output	4/4	4 - output produced according to instructions
self-evaluation	4/4	4 - self-evaluation accurate (or evaluating this item leads to an error)
total	100/100	